

```
*****Default*****
hostapd
Testing pico_w-mqtt
07/06/25
*****Default*****
```

You need several environment variables

```
export WIFI_SSID="WIFI-SSID"
export WIFI_PASSWORD="WIFI-PASSWORD"

export PIPELINEC_REPO="$HOME/PipelineC"
export PICO_EXTRAS_PATH=/home/devel/pico-extras

export PICO_SDK_PATH=/home/devel/pico2/pico-sdk
#export PICO_2350=1

export FREERTOS_KERNEL_PATH=/home/devel/pico2/freertos-lib/FreeRTOS-Kernel

export PICTOOL_FETCH_FROM_GITHUB=/home/devel/pico2/BUILT_PICOTOOL
```

```
devel@wifiextender:~ $ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.42.1 netmask 255.255.255.0 broadcast 192.168.42.255
    inet6 fe80::c831:9f7f:f5b9:4a5d prefixlen 64 scopeid 0x20<link>
    ether 00:0e:c6:8f:a2:6c txqueuelen 1000 (Ethernet)
    RX packets 88715 bytes 10862968 (10.3 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 187830 bytes 261245596 (249.1 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.235 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 2600:1700:69f0:42c0:5321:921c:c79a:7c64 prefixlen 64 scopeid 0x0<global>
    inet6 2600:1700:69f0:42c0::40 prefixlen 128 scopeid 0x0<global>
    inet6 fe80::f1ab:6da4:88d8:e3ed prefixlen 64 scopeid 0x20<link>
    ether b8:27:eb:2b:eb:c2 txqueuelen 1000 (Ethernet)
    RX packets 203163 bytes 261293049 (249.1 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 89075 bytes 12131497 (11.5 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
Contents of devel@pi5-90:~ $ ./scripts-rpi/route/2nd_level.sh
#!/bin/bash
route
sudo route add -net 192.168.42.0 netmask 255.255.255.0 gw 192.168.1.235 eth0
route
```

```
devel@pi5-90:~ $ ./scripts-rpi/route/2nd_level.sh
Kernel IP routing table
Destination   Gateway       Genmask      Flags Metric Ref  Use Iface
```

```

default      dsldevice.attlo 0.0.0.0      UG  100  0      0 eth0
192.168.1.0   0.0.0.0        255.255.255.0 U   100  0      0 eth0
Kernel IP routing table
Destination   Gateway         Genmask         Flags Metric Ref  Use Iface
default      dsldevice.attlo 0.0.0.0         UG   100  0    0 eth0
192.168.1.0   0.0.0.0        255.255.255.0 U   100  0    0 eth0
192.168.42.0  wifiextender    255.255.255.0 UG   0    0    0 eth0

```

```
devel@pi5-90:~ $ ./scripts-rpi/route/2nd_level.sh
```

```

devel@pi4-37:~ $ git clone https://github.com/develone/pico_w-mqtt.git -b dev-test-RP-FK
commit 7b82c43640a328a62888483db5cbb5112343520d (HEAD -> dev-test-RP-FK, origin/dev-
test-RP-FK)

```

```
Author: Edward Vidal Jr <develone@sbcglobal.net>
```

```
Date: Thu Jul 3 11:52:56 2025 -0600
```

need infor

```
devel@pi4-37:~ $ cd pico_w-mqtt
```

There are three scripts 1remote.sh, 3remotes.sh and 6remotes.sh

```
#!/bin/bash
```

```
./build_cli.sh
```

```
rm -rf remote5
```

```
mkdir remote5
```

```
cd remote5
```

```
cp ../exe-ocd.sh .
```

```
cmake -DPICO_BOARD=pico_w -DHOSTNAME="remote5" -DWIFI_SSID="WIFI-SSID" -
DWIFI_PASSWORD="WIFI_PASSWORD" ..
```

```
make
```

```
cd ../
```

```
echo "Done"
```

```
root@pi4-37:/home/devel# less /etc/hostapd/hostapd.conf
```

```
interface=wlan0
```

```
driver=nl80211
```

```
ssid=WIFI-SSID
```

```
hw_mode=g
```

```
channel=7
```

```
wmm_enabled=0
```

```
macaddr_acl=0
```

```
auth_algs=1
```

```
ignore_broadcast_ssid=0
```

```
wpa=2
wpa_passphrase=WIFI-PASSWORD
wpa_key_mgmt=WPA-PSK
wpa_pairwise=TKIP
rsn_pairwise=CCMP
```

```
devel@pi4-37:~/pico_w-mqtt/pi_tcp_tests $ ifconfig eth0
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.42.105 netmask 255.255.255.0 broadcast 192.168.42.255
    inet6 fe80::f749:56c8:dcef:6b0d prefixlen 64 scopeid 0x20<link>
    ether b8:27:eb:07:c3:34 txqueuelen 1000 (Ethernet)
    RX packets 1168 bytes 120195 (117.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1060 bytes 176184 (172.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
devel@pi4-37:~/pico_w-mqtt/pi_tcp_tests $ ifconfig wlan0
wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.32.1 netmask 255.255.255.0 broadcast 192.168.32.255
    inet6 fe80::b321:ad1f:1ecb:d9b2 prefixlen 64 scopeid 0x20<link>
    ether b8:27:eb:52:96:61 txqueuelen 1000 (Ethernet)
    RX packets 12273 bytes 504328 (492.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 25172 bytes 37424267 (35.6 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
devel@pi4-37:~ $ ./scripts-rpi/mosquitto/mos.sh
1751806446: mosquitto version 2.0.11 starting
1751806446: Config loaded from /etc/mosquitto/mosquitto.conf.
1751806446: Opening ipv4 listen socket on port 1883.
1751806446: Opening ipv6 listen socket on port 1883.
1751806446: mosquitto version 2.0.11 running
devel@pi4-37:~ $ 1751806447: New connection from 192.168.32.106:57277 on port 1883.
1751806447: New client connected from 192.168.32.106:57277 as remote5 (p2, c1, k10,
u'testuser').
1751806477: New connection from ::1:60742 on port 1883.
1751806477: New client connected from ::1:60742 as auto-93647E86-6EDF-1B7C-8D85-
315DFE4CB7E0 (p2, c1, k60, u'testuser').
1751806879: New connection from 192.168.42.105:45632 on port 1883.
1751806879: New client connected from 192.168.42.105:45632 as auto-7944AF4C-6C68-BC61-
DF2F-3A203CFC51FD (p2, c1, k60, u'testuser').
1751806879: Client auto-7944AF4C-6C68-BC61-DF2F-3A203CFC51FD disconnected.
1751806892: New connection from 192.168.32.106:63803 on port 1883.
1751806892: Client remote5 already connected, closing old connection.
1751806892: New client connected from 192.168.32.106:63803 as remote5 (p2, c1, k10,
u'testuser').
```

```
devel@pi4-37:~/pico_w-mqtt/pi_tcp_tests $ ./cli5
Socket created successfully
Connected with server successfully
Connected. iperf server 192.168.32.106 4001 mqtt_ip = 0x692aa8c0 mqtt_port = 1883 batt task
0x03b -> 0.047534 V 06/07/2025 13:01:33
```

6789012345678901234567890123456789012345678901230123456789012345678901
2345678901234567890123456789012345678901230123456789012345678901234567
8901234567890123456789012301234567890123456789012345678901234567890123
4567890123◆◆◆◆

```
devel@pi4-37:~/pico_w-mqtt/pi_tcp_tests $ iperf -c remote5
```

```
-----  
Client connecting to remote5, TCP port 5001  
TCP window size: 85.0 KByte (default)  
-----
```

```
[ 3] local 192.168.32.1 port 53534 connected with 192.168.32.106 port 5001  
[ ID] Interval      Transfer    Bandwidth  
[ 3] 0.0000-10.0782 sec 17.8 MBytes 14.8 Mbits/sec
```

```
devel@pi4-37:~ $ ./scripts-rpi/mosquitto/mos-q.sh  
T = 35.10 C 0x03b batt = 0.05 remote5  
T = 35.10 C 0x03b batt = 0.05 remote5  
T = 35.10 C 0x03b batt = 0.05 remote5  
T = 35.10 C 0x03b batt = 0.05 remote5  
T = 35.10 C 0x03a batt = 0.05 remote5
```